

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)		award (1) for either of following copper ... because there are no bubbles copper ... because there is no rise in temperature (accept no change in temperature) neutral answer – no reaction		1		1		1
		(ii)		51		1		1	1	1
	(b)			exothermic	1			1		
	(c)	(i)		award (2) for all points plotted correctly – tolerance $\pm\frac{1}{2}$ small square award (1) for 4 or 5 points plotted correctly award (1) for suitable curve from origin		3		3	3	
		(ii)		Graph stops at 60 s ☐ Graph is still rising at 60 s ☒ Graph reaches a maximum temperature of 56 °C ☐			1	1		

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		(iii)		The particles collide with less energy so less chance of successful collisions ☐ The particles move slower so less chance of successful collisions ☐ The particles have less surface area so less chance of successful collisions ☐ The particles get used up so less chance of successful collisions ☒	1			1		
		(iv)		increase the temperature (of the acid) (1) increase the concentration (of the acid) (1) accept use stronger acid neutral answer – use a catalyst / any reference to magnesium	2			2		2
		(v)		MgCl ₂		1		1		
				Question 2 total	4	6	1	11	4	4